INCH-POUND

MIL-M-48428A (AR)
15 October 1990
SUPERSEDING
MIL-M-48428 (MU)
15 October 1975

MILITARY SPECIFICATION

MICROWAVE CHASSIS ASSEMBLY: 10549047

This specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** **This specification covers the Microwave chassis Assembly, 10549047, hereinafter referred to as microwave chassis, which is a sub-assembly of the Receiver-Transmitter, RT-860/VPS-2, a part of Radar Set AN/VPS-2 (see 6.1).**

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army ARDEC, Attn: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressedStandardizationDocument appearing at the end of this document or by letter.

AMSC N/A FSC 1285

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MIL-R-50651 - Radar Set, AN/VPS-2:10549058
MIL-R-50668 - Receiver-Transmitter, Radar, RT-860/VPS-2: 10548089
MIL-D-50669 - Driver, switch: 10548602

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines
MIL-STD-2000 - standard Requirements for Soldering Electrical and Electronic Assemblies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the standardization Documents order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (see 6.4)

Us. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9342815 - Klystron Power Supply
10549047 - Microwave chassis Assembly
10548164 - IF Amplifier Assembly
10549058 - Radar Set AN/VPS-2
10549065 - Connector
10549173 - Lamp Holder
10549174 - Lens Indicator
10549175 - Oscillator Module
10549202 - Transmit Gate Assembly
10556081 - Connector
10558238 - Waveguide, Coupler
11729223 - Waveguide Adapter
11748124 - Radar System Tester, AN/VPM-2
11834450 - Klystron Beam Power Supply
9115C2039 - Connector

PACKAGING DATA SHEET

Ps 10549047 - Packaging of Microwave Chassis Assembly

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(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in **accordance with the applicable drawings and specifications.**

***3.2* Components and assemblies. The components and assemblies** shall comply with all requirements specified on Drawing 10549047, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the microwave chassis shall meet the requirements of MIL-F-13926 as follows:

Dimensions and tolerances
b. Inorganic protective surface finishings
c. Part identification and marking
d. workmanship

3.2.1 Subassembly. The subassembly of the microwave chassis shall have been inspected and accepted in accordance with the applicable military specification prior to assembling in the microwave chassis. The following subassembly is covered by the indicated specification:

UNIT NO.	SUBASSEMBLY	SPECIFICATION
2AlA8	Driver, Switch: 10548602	MIL-D-50669

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3.3 Environmental service conditions.

3.3.1 **Vibration. The microwave chassis shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to sinusoidal vibrations, in each of three mutually perpendicular axes, as follows:**

Frequency Range	Amplitude
10 to 15 hertz	**0.06 inches double amplitude**
15 to 25 hertz	**0.04 inches double amplitude**
25 to 40 hertz	**0.02 inches double amplitude**

The frequency of vibration shall be swept logarithmically.
The cycling time shall be 120 minutes per axis.
The sweep rate shall be 3.75 minutes ± 10% per decade.
The 10-40-10 hertz sweep time is 2 minutes 30 seconds ± 10%.

3.3.2 Shock. The microwave chassis shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± l0%, 11 milliseconds ± 10%, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude (non-operating) The microwave chassis shall meet the requirements of 3.4 and 3.5 after exposure for not less than 10 minutes to barometric pressure encountered at **40,000** feet **above** sea **level** (5.5 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The microwave chassis shall meet the requirements of 3.4, 3.5.1.1, 3.5.1.2, 3.5.1.5, 3.5.2.1, and 3.5.2.6 when stabilized thermally at any temperature between -40 degrees F' and +155 degrees F.

3.3.4.2 **Non-operating** (storage) . The microwave chassis shall not suffer permanent damage when subjected to the constant temperature extremes of -60 degrees F for a period of 24 hours and +160 degrees **F** for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating) The microwave chassis shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram of the microwave chassis.

3.5 Performance. The microwave chassis shall meet the following requirements with the inputs described in Table I.

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TABLE I. Input characteristics

REQUIREMENTS	CHARACTERISTICS	CONNECTION
Input power	+28.0 ± 0.1 Vdc +28 Vdc return -33.0 ± 0.1 Vdc -33 Vdc return ± 15 Vdc return +15.0 ± 0.1 Vdc -15.0 ± 0.1 vdc	J1-1, 3 J1-2, 4 J1-7 J1-8 J1-5 J1-6 J1-10
AGC control voltage	0.0 to 6.0 vdc	J1-A2
Diode switch driver trigger	**PRF 19.978 KHz** ± 200 **Hz** Pulse width 0.5 us Amplitude +18 V minimum into SO-ohm load with a SWR less than 1.15 to 1	J2
Modulation signal *	Gated 60 MHz burst PRF 19.978 KHz ± 200 Hz Width 280 ± 10 **us** (50% of peak) Amplitude 1.0 V pk-pk, minimum into 50-ohm load with a SWR of less than 1.15 to 1	**2A1A2J1**
Klystron power amplifier filaments **	-9.0 to -9.5 KVdc (Ref. beam power supply) 7.9 to 9.5 ma 5.0 to 5.8 vdc 2.0 to 2.8 Amps dc at a reference level of -9.0 to -9.5 KVdc	A3P4-B
	Return	**A3P4-C**

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TABLE I. Input characteristics (continued)

REQUIREMENTS	CHARACTERISTICS	CONNECTION
Klystron power amplifier grid ** (refer to figure 1)	Dc level -13.25 to -14.00 KVdc Pulse amplitude +4.25 KV to +5.00 **KV (Grid bias cut** off min. **4.25** KV) Rise time (10-90% points) 0.35 us maximum *** Pulse width 0.60 us to 0.80 **us** **PRF 19.978 KHz, ± 200 Hz**	A3P4-E
Klystron power amplifier collector	Ground	A3-E1

* Signal derived from the Transmit Gate Assembly 10549202.

** Signal derived from the Klystron Power Supply 9342815 and the Klystron Beam Power Supply 11834450.

***Limits are over temperature and production variations. Room temperature limits are 0.67 to 0.77 us.

3.5.1 Receive mode.

3.5.1.1 Overall gain. The overall gain of the microwave chassis, measured from the waveguide elbow (connected to HY1P2) to A5J2 with 0.0 Vdc at J1-A2, shall be 17 decibels (dB) minimum, 22 dB maximum, with an input level of -40 dBm.

3.5.1.2 AGC action. The gain of the microwave chassis shall be reduced by a minimum of 20 dB when the dc voltage at J1-A2 is changed from 0.0 Vdc to +6.0 Vdc. The gain versus AGC voltage curve shall be monotonic.

3.5.1.3 Noise figure. The noise figure for the microwave chassis shall be 12 dB maximum on all six channels. (11 dB maximum at room ambient).

3.5.1.4 Switch drive Pulse. The switch drive pulse at A7J1 shall exhibit the following parameters:

Negative amplitude:	22 Vdc minimum
Positive reference:	+0.7 Vdc minimum, +1.1 Vdc maximum
width: (50% points)	2.48 us minimum, 2.62 us maximum

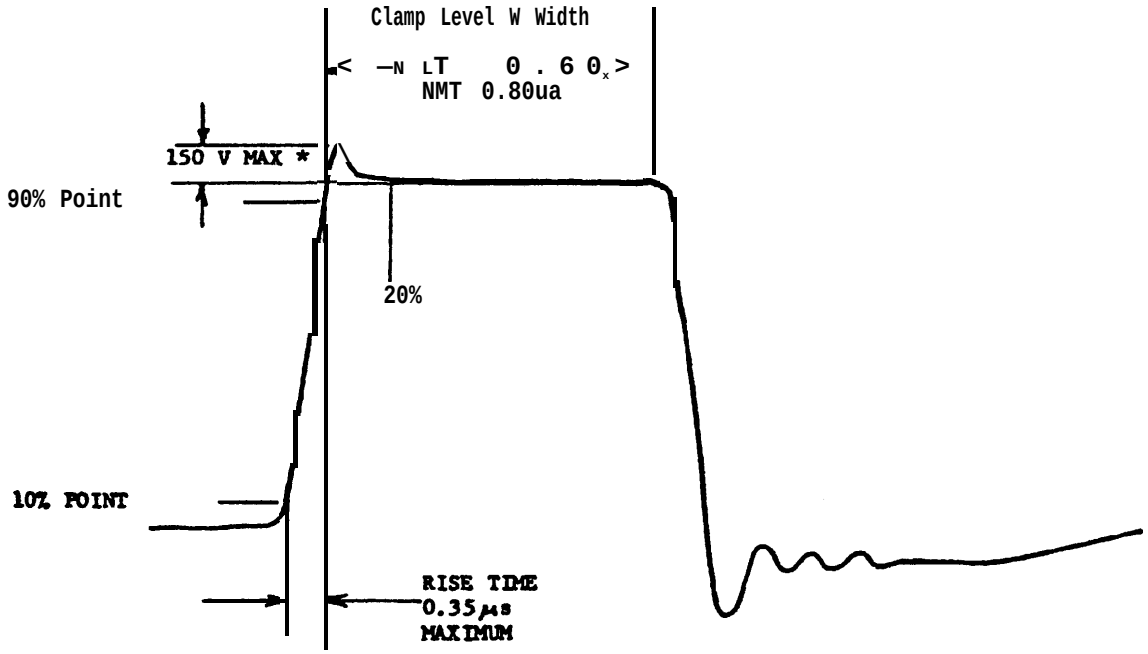
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3.5.1.5 Diode switch isolation. When the dc level at A7JI is changed from +1.0 to -33.o Vdc, the insertion loss of the diode switch shall increase a minimum of 45 dB.

3.5.1.6 Stamo oscillator test port. The microwave chassis shall have a waveguide Port that Provides a stamo oscillator frequency output at a power level-of not less than +3 dBm and greater than 20 dB isolation **from local oscillator. . .**

3.5.2 Transmit mode.

3.5.2.1 Average output power. The average output power, at PRF rate of 19.978 KHz ± 200 Hz and duty unit of .0054 minimum, as measured at the waveguide elbow [connected to HY1P2) shall be 4.26 watts (W), minimum for all channels.



* **RETURNING** TO 50 V MAXIMUM IN 20% of PULSE WIDTH.

FIGURE 1. Klystron power amplifier grid bias pulse (from klystron power supply).

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3.5.2.2 Rf drive adjustment. The microwave chassis shall have an attenuator that provides a means of varying the rf drive to the klystron power amplifier AlA3. Minimum attenuation shall be 0.3 dB maximum and the attenuation range shall be 28 dB minimum.

3.5.2.3 Rf power monitor. The microwave chassis shall have an indicator which when transmitting will provide a relative indication of the rf power being transmitted. There shall be an adjustment such that at nominal power output the indicator will read mid-scale.

3.5.2.4 Transmit indicator. The microwave chassis shall provide a switched ground path for an indicator lamp to 28 Vdc. This shall occur when in the transmit mode and the output power exceeds **1.80 ± 0.1 Watts (W).**

3.5.2.5 Transmit pulse envelope. The transmit pulse envelope, when detected and observed on an oscilloscope, shall have a cosine squared (COS^2) characteristic shape and shall be equal to the modulation signal pulse width +10, -O nanosecond (ns) at the 50-percent amplitude points, refer to figure 2.

3.5.2.6 Transmit frequency. The microwave chassis shall be capable of operation on six discrete frequencies by selecting one of six oscillator plug-in modules. The frequency of the transmitted signal shall be 9205 MHz ± 10 MHz using channel 1 oscillator module. Using the channel 1 measurements as a reference, the channel 2 through 6 output frequency shall be as follows:

Channel 2	Reference +8 ± 2 MHz
Channel 3	Reference +16 ± 2 MHz
Channel 4	Reference +24 ± 2 MHz
Channel 5	Reference +32 ± 2 MHz
Channel 6	Reference +40 ± 2 MHz

3.5.2.7 Output spectrum. The power of the 1st and 2nd order upper and lower side lobes of the transmit pulse shall be as follows:

SIDE LOBE	POWER BELOW MAIN LOBE
1st upper/lower	Not less than 26 dB
2nd upper/lower	Not less than 40 dB

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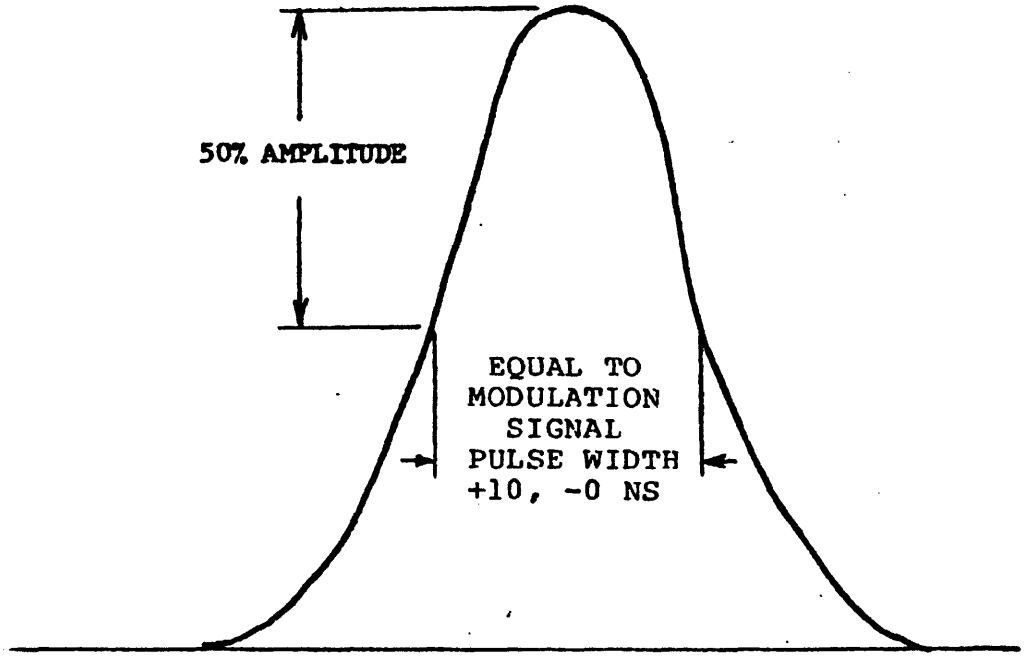


FIGURE 2. Transmit pulse envelope (detected rf).

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3.5.3 System performance.

3.5.3.1 Subclutter visibility. The microwave chassis shall meet the subclutter visibility requirements of 3.5.7 of MIL-R-50668 (AR) and 3.5.9.8 of MIL-R-50651 (AR).

3.6 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the general requirements of MIL-STD-2000 in addition to detail requirement task F of paragraph 1.2 of MIL-STD-2000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.1.1 **Responsibility for compliance.** **All items** must meet all requirements of sections 3 and 5 and all applicable drawings. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance **comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.**

4.1.2 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification.

4.2 Classification of inspections. The inspection **require-**ments specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Special sampling inspection (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50 deg to +90 deg F.
b.	Relative humidity	90% RH Max.
c.	Pressure	28 to 32 inches of Hg.

4.3 First article inspection.

4.3.1 First article submission. The contractor shall submit a first article sample as designated by the contracting officer consisting of three microwave chassis for inspection in accordance with 4.3.2.

4.3.2 Inspections to be performed. First article assemblies, components and test specimens may be subjected by the Government to any or all of the examinations and tests specified in Table II and Table III herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly, component or test specimen to comply with any of the stated requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies. All components, assemblies and subassemblies shall be inspected in accordance with the inspection provisions contained in this

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specification and the **Quality** Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with the sampling plans specified in 4.4.2.3 herein. Examination and test for packaging and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in 4.4.2.3 shall constitute the minimum inspection to be performed by the contractor after first article approval and prior to Government acceptance or rejection by item or lot.

4.4.2.3 Functional tests. The tests in Table II shall be performed on a 100% basis.

TABLE II. Functional tests
(100% Inspection)

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
101.	**Electrical**	3.4	4.7.2
102.	Performance	3.5	4.7.3
Minor:		**None**	**N/A**

4.5 Special sampling. One microwave chassis out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests of Table II and shall then meet the tests in Table III. Unless otherwise specified in the contract the items shall be returned to the lot upon successful completion of testing.

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TABLE III. Special sampling

CHARACTERISTIC		REQUIREMENT	TEST PROCEDURE
Critical:		None	N/A
Major:			
103.	Vibration	3.3.1	4.7.1.1
104.	Shock	3.3.2	4.7.1.2
105.	Altitude	3.3.3	4.7.1.3
106.	Temperature	3.3.4	4.7.1.4
107.	Humidity	3.3.5	4.7.1.5
Minor:		None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be-suspended by the Government until necessary corrections to all production since the last test as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, **SUpply . and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.**

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with the design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 whenever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which

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does not exceed 90 percent of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the test equipment listed in Table IV shall be used in the performance of the applicable test methods and procedures as specified in 4.7.

TABLE IV. Test equipment list

ITEM NUMBER	DESCRIPTION
1. Rf load	Bandwidth: 9150 to 9250 MHz minimum Average power rating: 10 w minimum. VSWR: 1.0:1.1 maximum.
2. Crystal detector	Bandwidth: 9150 to 9250 MHz minimum. Average power rating: 10 mW minimum. VSWR: 1.0:1.8 maximum with 100-ohm load and 0.5 mA crystal current.
3. Directional coupler (2 required)	Bandwidth: 9150 to 9250 MHz minimum Peak power rating: 2000 w minimum. coupling: 40 dB ± 0.9 dB. Mainline VSWR: 1.0:1.1 maximum. Secondary line VSWR: 1.0:1.2 maximum.
4. Signal generator	Frequency range 9150 MHz to 9250 MHz minimum. Frequency stability: with temperature less than 0.006%/ deg C change in ambient temperature; with line voltage - less than 0.02% change for line voltage variation of ± 10%. Residual FM; less than 15 KHz peak. Output range: 0.224 V to 0.1 UV rms, into a 50-ohm load. Output calibration: 0 dBm to -127 dBm. Source impedance: 50-ohm nominal.
5. Frequency meter	Type; direct reading resonant cavity. Frequency range: 9150 to 9250 MHz minimum. Overall accuracy: 0.08%.
6. Thermistor mount	Frequency range: 9150 to 9250 MHz minimum. Maximum average input power: 10mW minimum.

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TABLE IV. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
7.	Power meter	Frequency range: 9150 to 9250 MHz minimum. Power range: lmW full scale. Accuracy: +1% of full scale. Type: Thermistor sensing element.
8.	I-f amplifier	10548164, or equivalent.
9.	Noise figure meter	Type: automatic readout with argon tube waveguide noise source and 60 MHz integral i-f amplifier. Scale: not less than 20 dB full scale; 12 dB calibration point shall be not less than half scale.
10.	Waveguide noise source	Type: argon-filled gas-discharge tube in standard waveguide house. Output connections: UG-39/U. Excess noise rating: 15.75 ± 0.5 dB.
11.	Oscilloscope	Vertical display: dual trace, dc to 75 MHz (approximately 3 dB down) at 10 mV/cm minimum. Horizontal sweep: calibrated time base 50 ns/cm to 1 s/cm minimum. Delayed sweep: 2 us to 50 us, continuously variable, minimum.
12.	Power supply	Type: laboratory supply. Minimum requirements: 0 to 32 Vdc. Load regulation: 0.02%. Load current capability: 15A. Ripple and noise: l mV rms or 5 mV peak-to-peak maximum from dc to 20 MHz.
13.	Power supply	Type: laboratory supply, independent dual voltage source. Minimum requirements: O to 20 vdc. Load regulation: 0.02%. Load current capability: loom. Ripple and noise: lmv rms or 5 mv peak-to-peak maximum for dc to 20 MHz.

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TABLE IV. Test equipment list (continued)

ITEM NUMBER		DESCRIPTION
14.	Power supply	Type: laboratory supply. Minimum requirements: O to 40 Vdc. Load regulation: 0.02%. Load current capability: loom. Ripple and noise: lmV rms or 5 mv peak-to-peak maximum for dc to 20 MHz.
15.	Rf voltmeter	Ranges: 3mV to 300 mV in four ranges. Frequency range: 1 MHz to 100 MHz minimum. Accuracy: ± 3% of full scale.
16.	Digital voltmeter	Display: four digit readout. Accuracy: 0.01%. Range: 100 Vdc.
17.	Pulse generator	Pulse amplitude: +18 V peak. Pulse width: 0.5 us. Rise time: 100 ns maximum.
18.	Test fixture	See figure 3, and table v.
19.	Rf attenuFrequency range:	dc to 100 MHz. Attenuation: O to 30 dB in l dB steps.
20.	Transmit gate assembly	10549202, Subassembly of Receiver-Transmitter, RT-860\VPS-2.
21.	Klystron power supply assembly	9342815, subassembly of Receiver-Transmitter, RT-860A/PS-2.
22.	Klystron beam power supply	11834450, Subassembly of Receiver-Transmitter, RT-860/VPS-2 .
23.	Rf signal generator	Frequency: 59.936 MHz. Output level: 0.1 uV to 3.0 V rms into 50-ohms. Source impedance: 50-ohms nominal. Residual FM: Less than ± 1 part in 10^6 or ± 20 Hz peak, whichever is greater.

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TABLE IV. **Test equipment list (continued)**

ITEM NUMBER		DESCRIPTION
24.	Spectrum analyzer	Type: Laboratory. Frequency range: 10 MHz to 12 GHz. Sensitivity: -05 dBm. Rf input attenuation: 0-60 dB. Spectrum width: 100 KHz - 2 GHz. Vertical display: Logarithmic 10 dB div.
25.	Waveguide to "N" (female) adapter	Frequency range: 8.20 to 12.0 GHz. SWR:NMT 1.25 to 1.
26.	Waveguide, coupler	10558238
27.	Coaxial termination	Impedance: 50-ohms. VSWR : 1.15 to 1.
28.	Oscillator Modules Channel 1-6	10549175 - Oscillator Module
29.	Radar System Tester AN/vPM-2	11748124, or equivalent
30.	Radar Set AN/VPS-2	10549058

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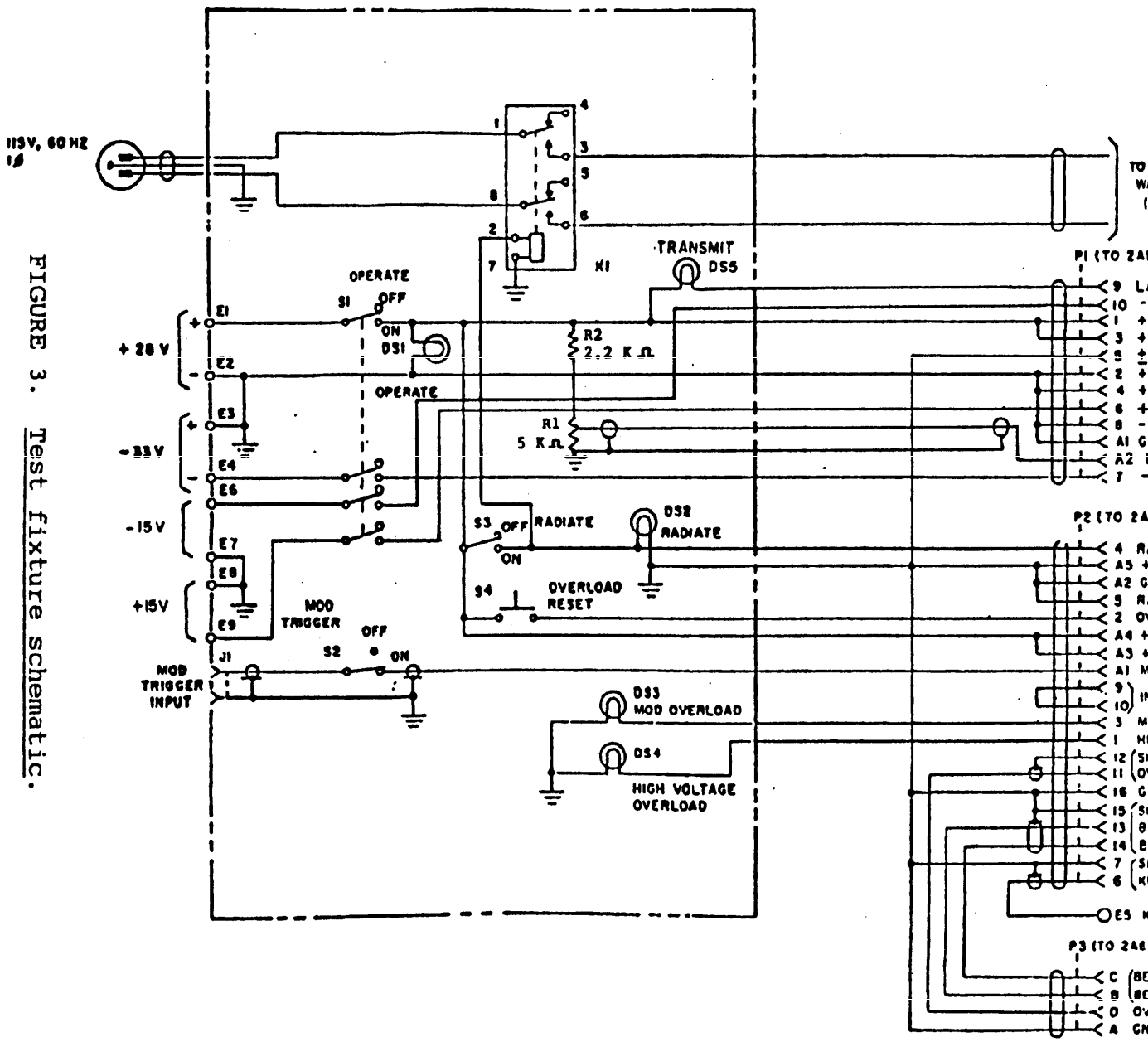


FIGURE 3. Test fixture schematic.

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TABLE v. Test fixture parts list

DESIGNATION	PART	DESCRIPTION
DS1 – DS4	Lamp	MS25231-1819 (28 Vdc, 0.04A)
DS 5	Lamp	MS25231-313 (28 Vdc, 0.17A)
El - E4 E6 - E9	Binding post	Superior, DF30, or equivalent
E5	Terminal lug	No. 8 size
J1	Connector	UG-625/U
K1	Relay	Magnecraft, W88CPX-16, or equivalent
MP1	Chassis	Bud, AC-404, or equivalent
P1	Connector	10556081
P2	Connector	10549065
P3	Connector	9115C2039
S1	Switch	4PDT, Cutler Hammer, 7691K14, or equivalent
S2	switch	SPST, Cutler Hammer, 8282K14, or equivalent
S3	Switch	SPNO, Cutler Hammer, 8803K6, or equivalent
W1	Line cord	3 wire, 115 Vat.
XDS1-XDS4	Lamp holder	Dialco, 95-0408-0931-241, or equivalent
XDS5	Lamp holder	10549173
	Lens, indicator	10549174-2
XK1	Relay socket	Amphenol, 78RS-8, or equivalent
R1	Potentiometer	0-5 K OhmS, 2W
R2	Resistor	2.2 K OhmS, 5%, l/2w

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810 except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. The test shall be performed as specified in Method 514, Category 8, Procedure I, of MIL-STD-810 as modified by 3.3.1 herein. In Step 1 of Procedure I only steps 4 through 8 of paragraph II-2.1 apply. The microwave chassis shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the microwave chassis shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock . This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's and 11 ms ± 10% duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10%.

d. The shock measuring system shall have a flat frequency response within ± 10% from 20 to 2,000 hertz.

After the application of all the shock impulses, the microwave chassis shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (non-operating). This test shall be performed as specified in Procedure III, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational in chamber checkout shall be as specified in 4.7.2, 4.7.3.1, 4.7.3.2, 4.7.3.5, 4.7.3.7, and 4.7.3.12 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational in chamber checkout shall be as specified in

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4.7.2, 4.7.3.1, 4.7.3.2, 4.7.3.5, 4.7.3.7, and 4.7.3.12 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating) This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except **that in Step 7, checkout shall be performed at the conclusion of-the test only, with the item at standard ambient conditions. The checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.**

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment (SME) and visual examination for compliance with 3.4. All components and subassemblies shall have been electrically checked for compliance to their individual requirements at a lower level.

4.7.3 Performance. Connect the microwave chassis in the test set-up shown in figure 4, and adjust the test equipment as required in table VI.

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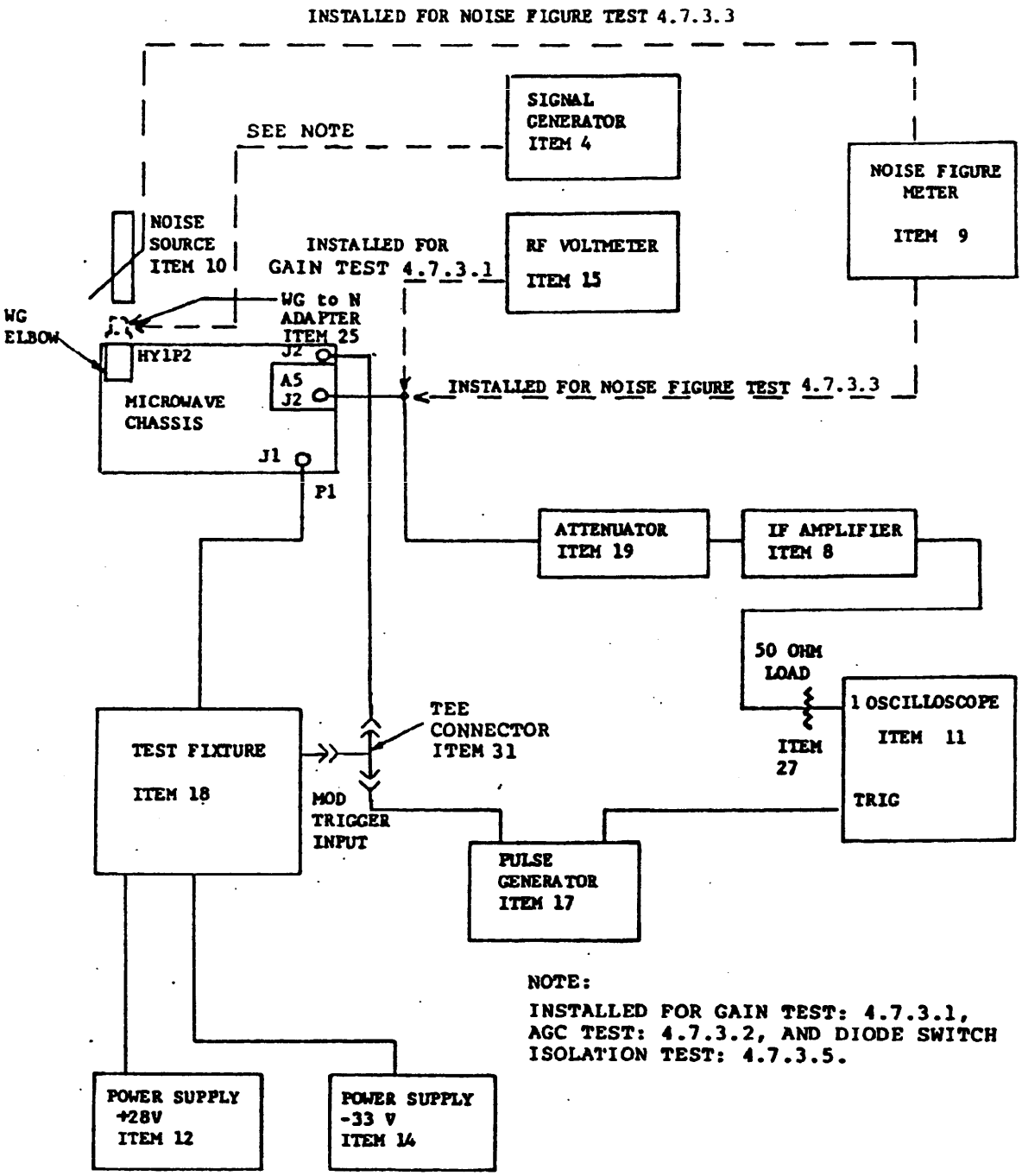


FIGURE 4. Test set up for receive operation.

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TABLE VI. Test equipment adjustments

TEST EQUIPMENT	ADJUST FOR
Pulse generator, item 17.	**PRF : 19,978 ± 200 Hz** Pulse width: 0.5 us. pulse amplitude: +18 V **(minimum)**
Power supply, item 12.	+28 Vdc ± 0.1.
Power supply, item 14.	***-33*** vdc +0.1.

4.7.3.1 Overall gain.

Connect the signal generator (item A, figure A) output to the waveguide elbow (connected to HY1P2) input/output port of the microwave chassis, using the waveguide to N adapter (item 25, figure 4). Set the output of the signal generator to -4o dBm.

b. Connect the rf voltmeter (item 15, figure *4)* to **A5J2 mixer pre-amplifier.**

Plug channel 4 crystal module into the x-band source (Al); tune the signal generator for a peak indication on the rf voltmeter. Set the AGC voltage control, Rl, on the test fixture for 0.0 Vdc at J1-A2. The rf voltmeter shall indicate not less than -23 dBm and not more than -18 dBm. Verify compliance to 3.5.1.1.

4.7.3.2 AGC action

a. Disconnect the rf voltmeter and connect A5J2 to the attenuator (item 19, figure 4), i-f amplifier (item 8, figure 4) and oscilloscope.

b. Set 20 dB attenuation in the attenuator and establish a reference level on the oscilloscope, peaking up the signal generator tuning if necessary.

C. **Advance the AGC voltage, using R1 on the test fixture, from 0.0 to +6.0 V at the same time reducing the attenuator to maintain reference level established in b. At 6.0 v AGC and O dB attenuation, the signal output shall not exceed the reference level to meet the requirements of 3.5.1.2.**

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4.7.3.3 Noise figure.

a. Connect the waveguide noise source (item 10, figure 4) to the waveguide elbow (connected to HY1P2) and connect the noise figure meter (item 9, figure 4) to the waveguide noise source.

b. Connect a 50-ohm coaxial test cable from A5J2 to the i-f input on the noise figure meter.

c. Measure noise figure using the procedure given in the manufacturer's manual for the noise figure meter - reading shall be not more than 12 dB (11 dB maximum at room ambient) .

d. Repeat 4.7.3.3.3, installing each of the other five crystal modules, in turn, in the microwave assembly. The reading on each channel shall be not more than 12 dB (11 dB maximum at room ambient) to meet the requirement of 3.5.1.3. Disconnect the noise figure test equipment.

4.7.3.4 Switch drive pulse. Connect a test cable to 2A1A7J1 and 2P5, using a tee connector, and connect the other end of the cable to the oscilloscope channel 2 (item 11, figure 4). The switch drive pulse parameters shall meet the requirements of 3.5.1.4. Remove the test cable and tee connector and reconnect 2P5 to 2A1A7J1.

4.7.3.5 Diode switch isolation.

a. Connect the signal general (item 4) to the waveguide elbow (connected to HY1P2) input port. Connect a test cable from preamp i-f output connector on the test fixture to channel 2 input on the oscilloscope with a 50-ohm load on the oscilloscope end.

b. Set the signal generator output level to -20 dBm and peak the frequency for maximum deflection on the oscilloscope.

c. Adjust the oscilloscope delayed sweep for a 5 us time base synchronous with pulse generator trigger (item 17, figure 4).

d. With the display as shown in figure 5, verify that the rf level during the gated off time is 45 dB or greater than the level during the on time. Verify compliance to 3.5.1.5.

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4.7.3.6 Stamo oscillator test port. Connect the power meter (item 7), using waveguide, coupler (item 26) and waveguide to "N" adapter (item 25), to the stamo oscillator test port of the microwave chassis. The power meter indication shall not be less than +3 dBm. Verify compliance to 3.5.1.6.

4.7.3.7 Average output power.

a. Connect the microwave chassis in the test set-up as shown in figure 6.

b. Turn on switch s3 of the test fixture and tune the klystron for **maximum** output at the operating frequency. The power meter (item 7, **figure 6) shall indicate not less than 0.426** mW. (Note: With 40 dB coupler, 0.426 mw at the coupled port indicates 1 kw peak pulse **output at through port.)**

c. **Repeat 4.7.3.7.2 on each of the five channels to meet the requirements of 3.5.2.1.**

4.7.3.8 **RF drive adjustment.** Remove the cap from AT1 and loosen lockscrew. Turning the adjustment screw at AT1 shall vary the average power by more than **1.0** dB. Reset to nominal power level and tighten lockscrew, replace cap. Verify compliahce to 3.5.2.2.

4.7.3.9 RF Power monitor. With a nominal 5 watt average power output (0.5 milliwatt on power meter), adjust resistor AIR1 so that the RF Power Monitor AIM1 reads mid-scale. Verify compliance to 3.5.2.3.

4.7.3.10 Transmit indicator illumination threshold.

a. Turn on test fixture switch S3. Using power meter (item 7, figure 6) measure and note the average transmitter power output.

b. Remove cap of attenuator AT1 and loosen lockscrew.Adjust attenuator AT1 for transmitter output of 180 ±10 milliwatts (mW).

c. Adjust transmit indicator threshold potentiometer A9R2 SO that the transmit indicator DS5 on the test fixture -just illuminates.

d. Readjust attenuator AT1 until transmitter power output is at the original level noted in 4.7.3.10.1. Lock attenuator lockscrew and replace protective cap. Turn off test fixture switch s3.

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4.7.3.11 Transmit pulse envelope. With the **microwave chassis** in radiate condition, observe the detected rf envelope on channel 1 **of the oscilloscope (item 11).** Check that the detected rf envelope (COS^2 pulse) is equal to modulation signal or NMT 10 ns wider at the 50 percent amplitude points, refer to figure 2, to meet the requirements of 3.5.2.5.

4.7.3.12 Transmit frequency.

a. With **test fixture switch s3 on, observe the detected rf** pulse display, tune the cavity frequency meter (item 5, figure 6) for a slight dip in pulse amplitude. At the point of maximum dip, read the frequency from the scale of the frequency meter. The reading shall meet the requirements of 3.5.2.6.

b. Repeat 4.7.3.12.1 with each of the other five crystal modules installed, tuning the klystron to each operating frequency to verify their compliance to 3.5.2.6.

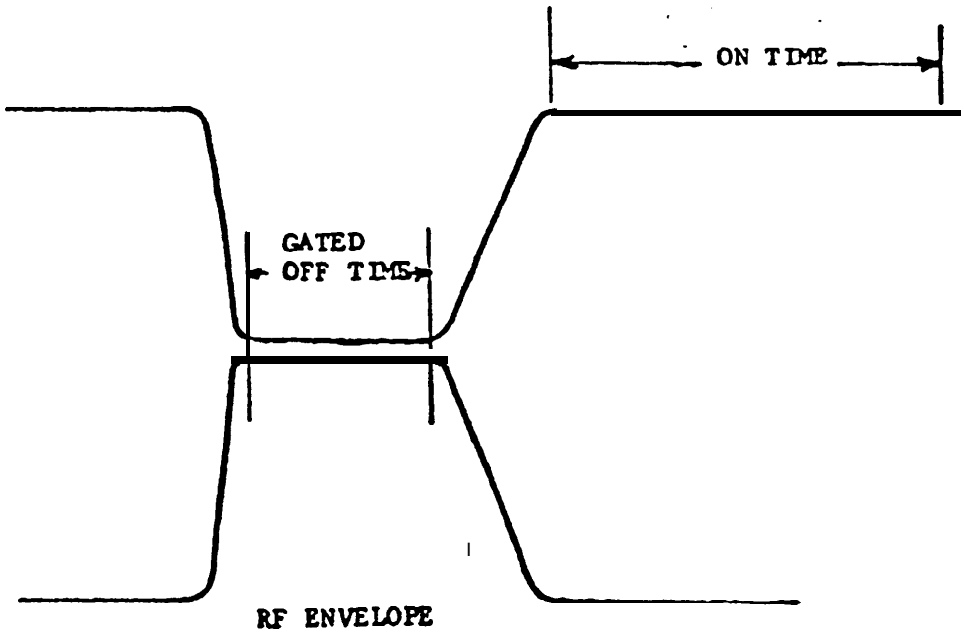


FIGURE 5. Diode switch isolation pattern.

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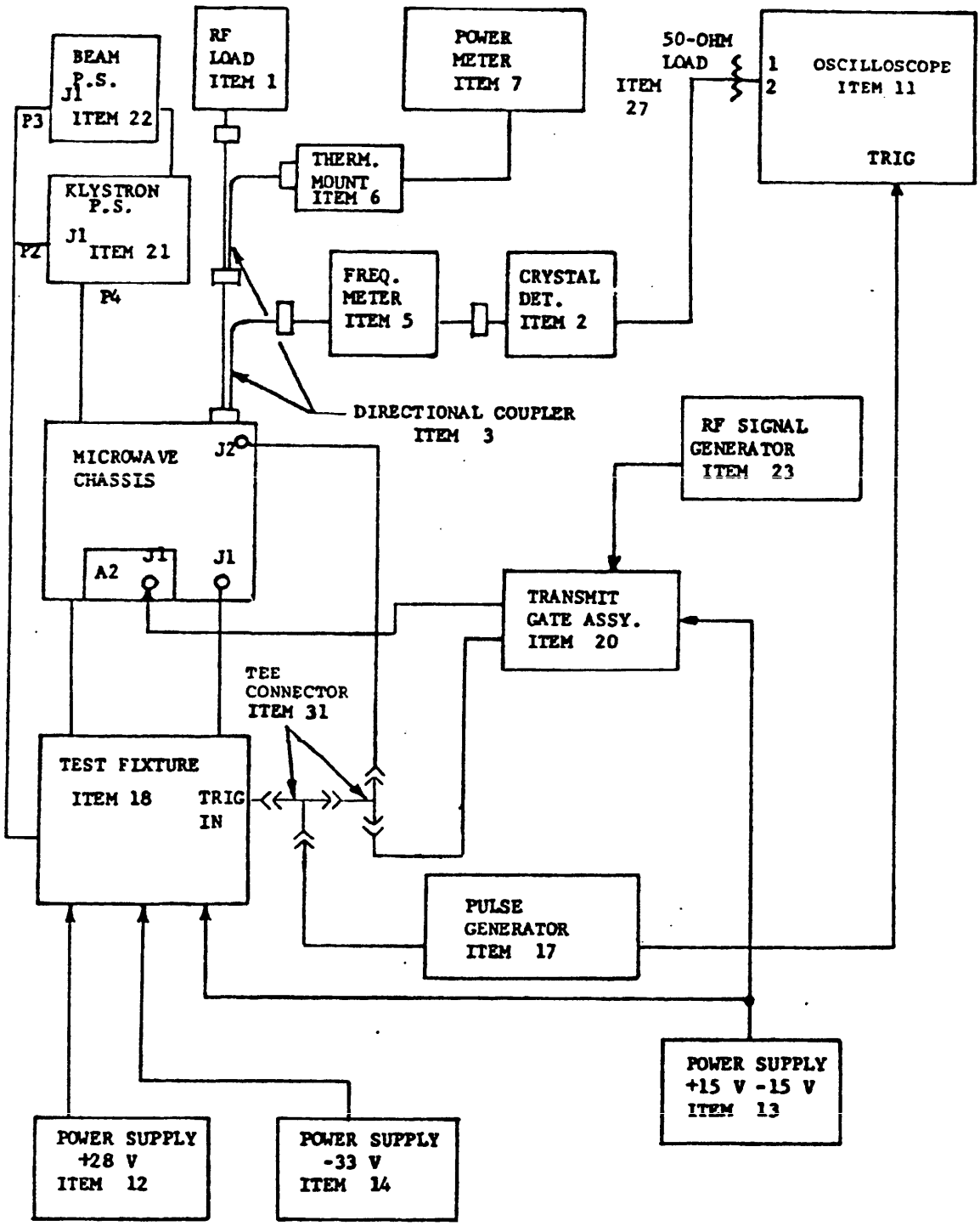


FIGURE 6. Test set up for transmit operation.

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4.7.3.13 Output spectrum.

Disconnect the crystal detector (item 2, figure 6) from the frequency meter (item 5, figure 6). Using an X-band test cable, connect the input of the spectrum analyzer (item 24, figure 6) to the frequency meter.

b. With test fixture switch S3 on, adjust the spectrum analyzer for a proper display. Verify the side lobes suppression compliance to requirements of 3.5.2.7.

4.7.3.14 Subclutter visibility. Using the Radar System Tester (item 29) the microwave chassis when installed in the Radar Set AN/VPS-2 shall meet the subclutter visibility requirements of paragraphs 3.5.7 of MIL-R-50668 and 3.5.9.8 of MIL-R-50651.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of **MIL-STD-2000 as specified in paragraph 3.7.**

S. **PACKAGING**

5.1 General. Preservation, packing, and marking shall **be** in **accordance with Packaging Data Sheet PS 10549047.**

6. (This section contains information of a general or explanatory **nature that may be helpful, but is not mandatory.)**

6.1 Intended use. The microwave chassis is a subassembly of the Receiver-Transmitter, RT-860/VPS-2, a part of Radar set AN/vPs-2. The receiver-transmitter provides sufficient pulsed rf power to obtain usable echo return from targets of interest and converts these target signals to intermediate frequency for use in other units of the system.

6.2 Acquisition requirements. Procurement documents shall specify the following:

6.2.1 Procurement requirements.

Title, number and date of this specification.
b. Applicable stock number.
c* First Article Sample requirements.
d. Packaging requirements, if other than specified **in** Section 5.
e. Serialization requirements, if applicable.

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6.3 Submission of contractor inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army AMCCOM, ATTN: AMSMC-QAF-I , Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-OT-90-12009 will apply.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research, Development and Engineering Center (ARDEC), may also include drawings prepared by, and identified as, U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now **under cognizance of ARDEC.**

6.5 Subject term (key word) listing.

Microwave chassis
Receiver-Transmitter, RT-860/VPS-2
Radar Set AN/VPS-2
Vulcan Air Defence System (VADS)
Fire Control System

6.6 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A712)

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